Insurance Pricing with GLMs

1. Overview

In this project, we are interested in modeling claim frequency and severity using the features given in the dataset, such as driver age, policy region, and area (rural, urban, etc.). These models will be used to estimate the pure premium and provide business insights for various policies.

2. Data and Data Limitations

This analysis draws on one data source:

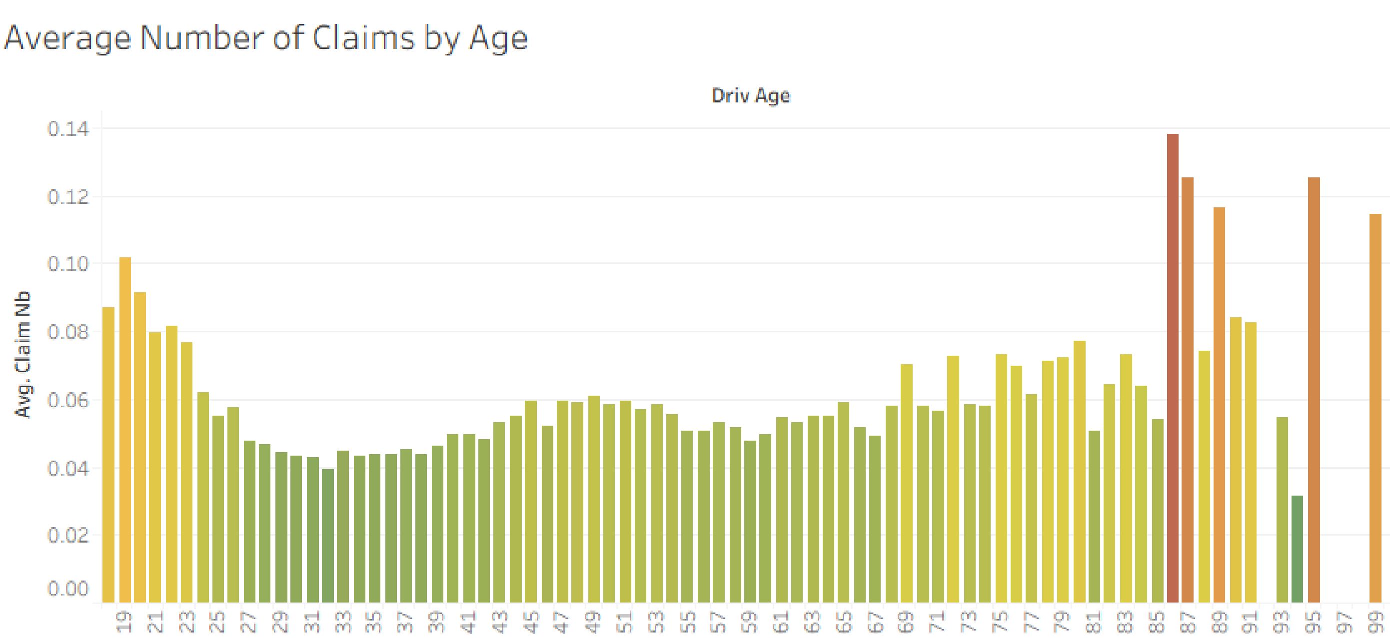
1. French Motor Third-Party Liability dataset:
    - freMTPL2freq (frequency dataset)
    - freMTPL2sev (severity dataset, linked to frequency dataset via "IDpol" column)

The dataset was validated using R. The dataset was complete (no missing or invalid values), so no data cleaning was necessary to begin analysis. Furthermore, there was a sufficient sample size for all features (678,013 observations for frequency, 26639 observations for severity), which implies that the data is credible.

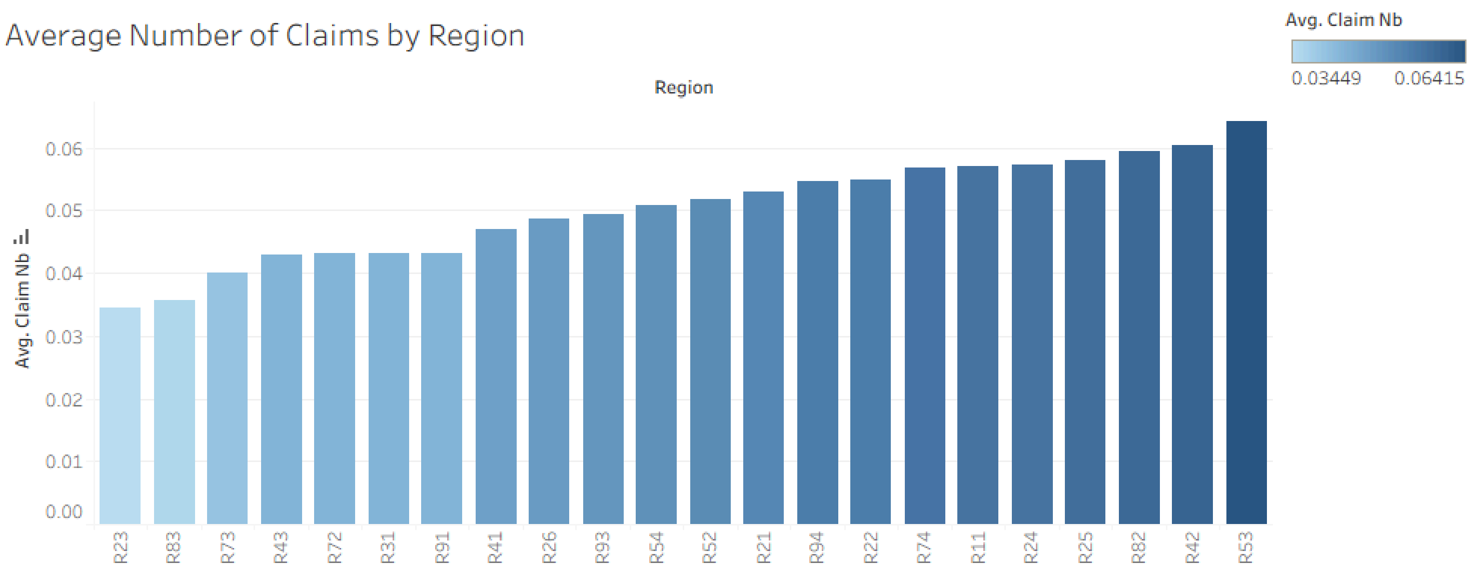
3. Claim Frequency

A preliminary analysis of the claim frequency data indicates that multiple features are relevant for predicting claim count. Specifically, driver age, policy region, "bonus malus", and area seem particularly useful.

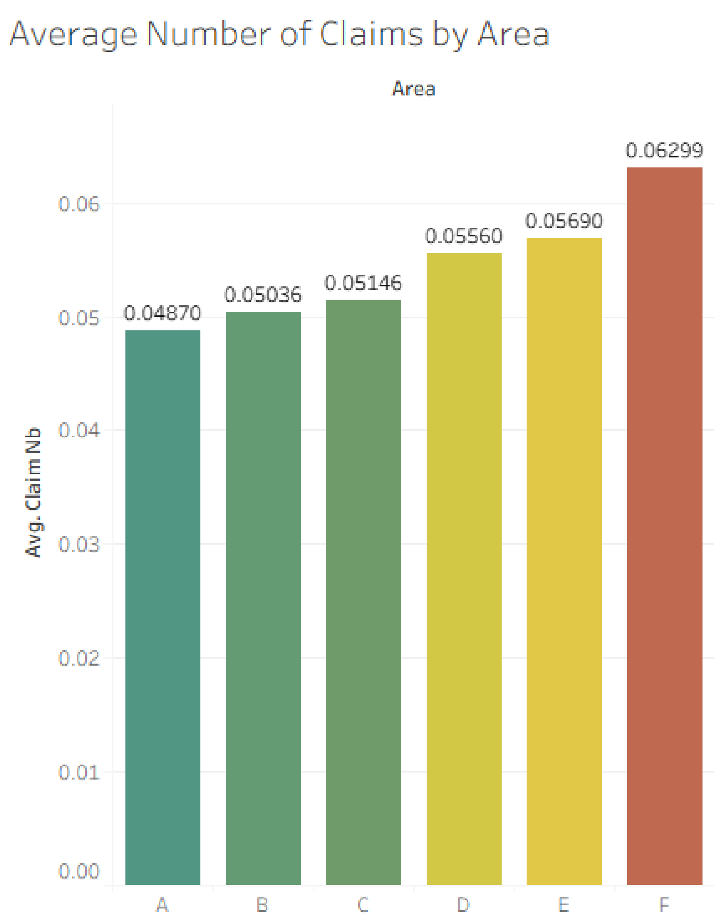
For driver age, it is apparent that both younger and older drivers exhibit higher average frequency of claims, with older drivers (86+) having the highest average frequency of claims.



Policy region refers to the policy region in France, based on the 1970 - 2015 classification). For policy regions, there is a wide range of average frequencies across regions ~ (0.035 - 0.064), with some regions having higher frequency, and others having lower frequency.

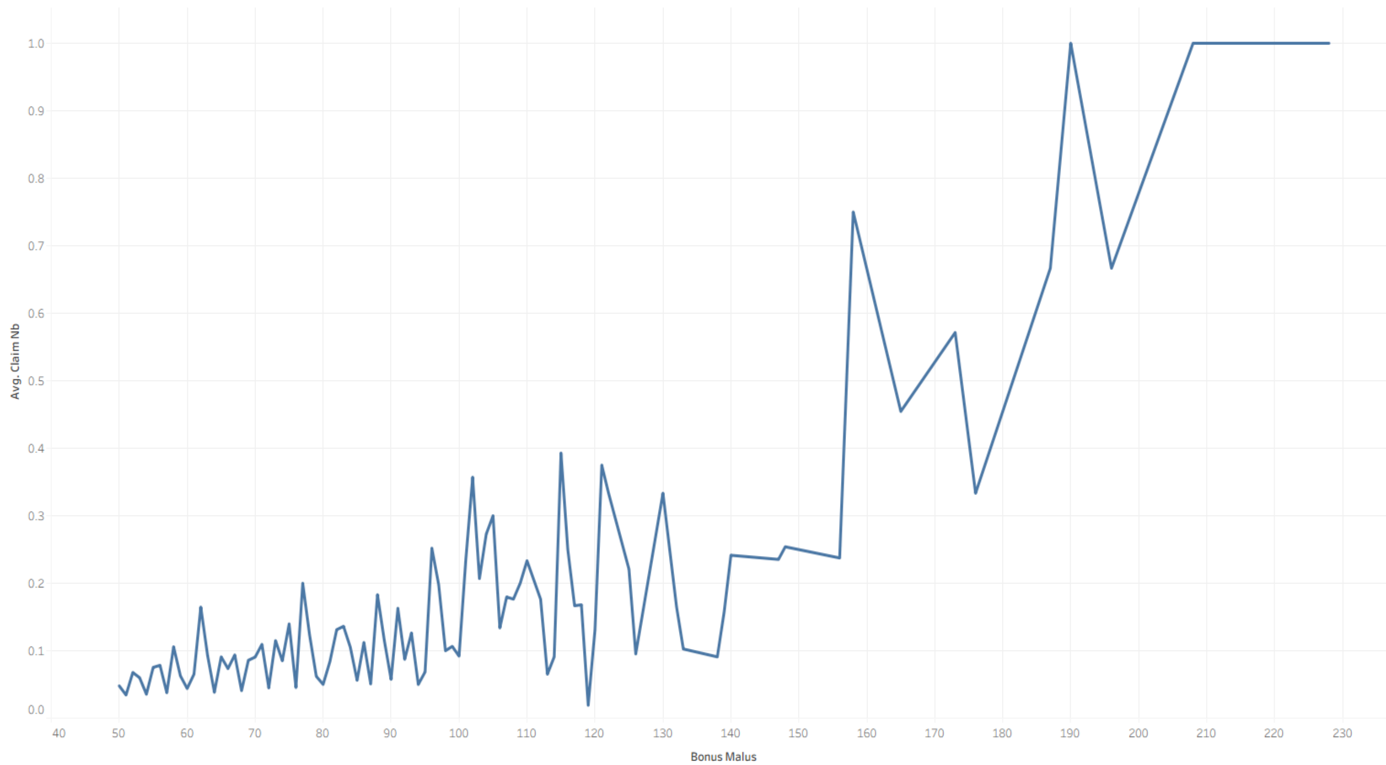


Area refers to the density value of the city community where the driver resides ("A" for rural - "F" for urban center). For area, a similar result is observed as for policy region. Intuitively, areas with higher population density (D, E, F) exhibit higher average claim frequency.



"Bonus Malus" is an insurance mechanism that changes value depending on claim history. In this dataset, it takes on values from 50 to 350, with a value < 100 representing a history of less claims, and a value > 100 being a history of more claims. Unsurprisingly, a history of higher claims appears associated with higher average claims in the present.

Average Number of Claims vs. Bonus Malus



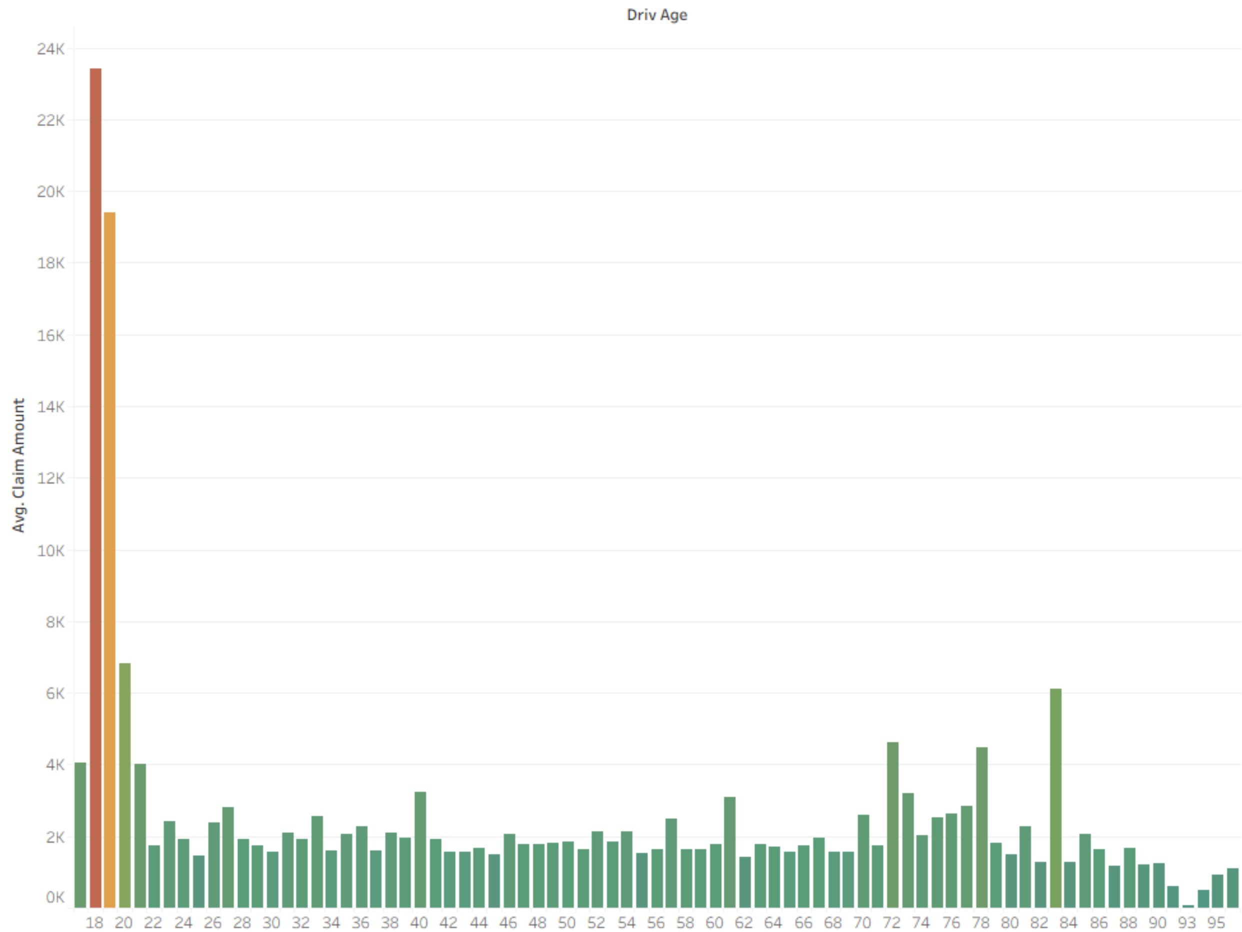
The trend of average of Claim Nb for Bonus Malus. The view is filtered on average of Claim Nb, which ranges from 0.0010 to 1.0000.

For modeling claim count, it is necessary to use a Generalized Linear Model (GLM) to model count data. Since the mean and variance are similar for claim count (mean = 0.05325, variance = 0.05766), a Poisson GLM with a log link is a suitable candidate, adding a log transformation of exposure to account for the fact that policies have been in effect for varying time lengths.

The frequency model incorporates all predictors that are observable before the policy is written (driver age, area, vehicle power, vehicle age, bonus/malus, vehicle brand, vehicle gas type, population density, and region). A summary of this model can be found in the appendix at the end of this document. The z-tests confirm the previous observations. All predictors except for population density are all significant predictors of claim count. Population density is removed from the model, since the model that excludes it slightly outperforms the full model. Notably, the overall model explains very little of the overall variation in claim count, with an AIC value of 286,701. We can also expect the model to have slightly higher test error on younger drivers, since the model implies that expected claim count strictly increases with age, which contradicts what was observed previously.

4. Claim Severity

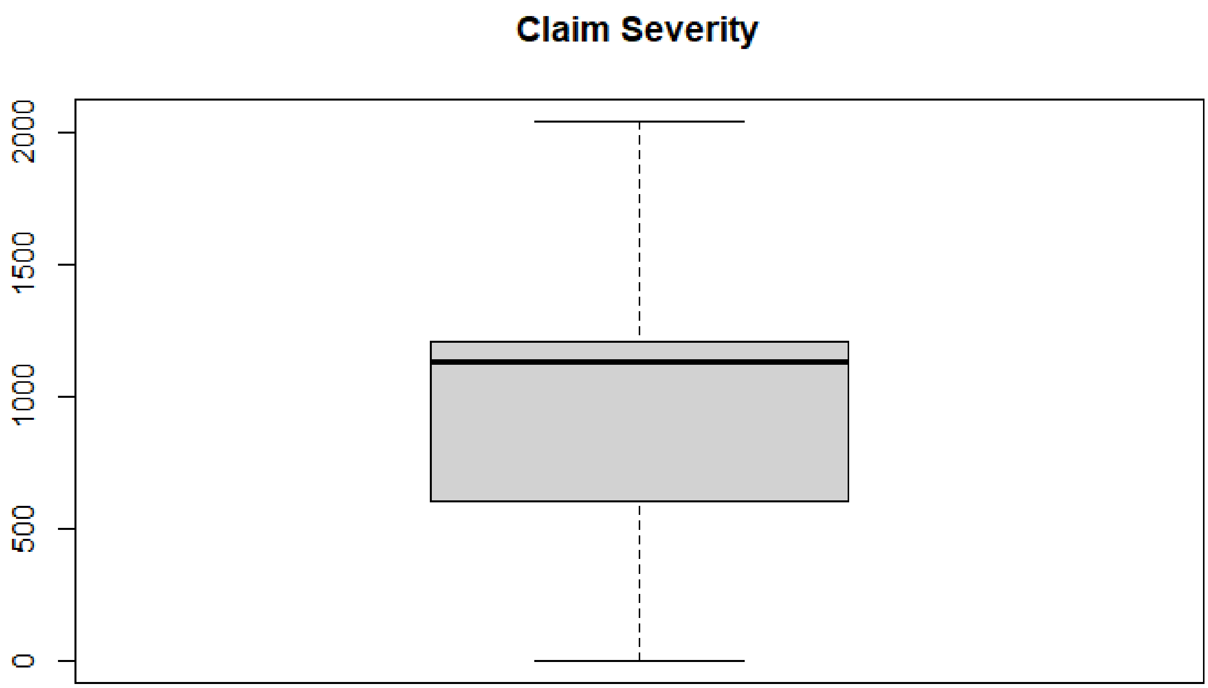
For claim severity, there are less significant predictors. For example, when plotting driver age against average claim amount, it appears that drivers age 18 or 19 have significantly elevated average claim amount. However, these elevations are due to two abnormally large claim amounts in the dataset ($1,301,172.60 and $4,075,400.56). These values should be further investigated to determine their legitimacy.



When observing average claim amounts against all other predictors in the dataset, few observable patterns are present. Bonus/malus is seemingly the best predictor of claim severity, exhibiting a slight upwards trend.

Although many of the predictors do not seem significant, they will still be included in the model to determine their statistical significance as well as assess if they contribute at all to predicting claim severity.

The distribution of claim amount is right-skewed as shown in the boxplot below. Therefore, a gamma GLM is a suitable candidate for modeling claim severity. Compared to a Log Normal GLM, the gamma model has a lighter tail (less very large values), which better reflects the distribution of claim severity.



The selected model is the model with the lowest AIC of 455,576. This is the full model including all predictors except for the number of claims. This variable was excluded because it is assumed that claim frequency and severity are independent of each other, as discussed in section 6. The statistically insignificant predictors remain in the model, since the goal is predictive accuracy. A summary of this model can be found in the appendix at the end of this document.

5. Pure Premium Estimate

With models selected for frequency and severity, the pure premium estimate can now be calculated as:

$$\text{Pure Premium} = \text{Expected Claim Frequency} \times \text{Expected Claim Severity}$$

Thus, for a given set of characteristics (observable before the policy is written), we can estimate the expected loss cost. For example, for a 25 year-old driver from region R21 with vehicle brand B10, vehicle age of 3 years, regular gas, vehicle power of 4, population density of 700, in area type A with and bonus/malus score of 50:

- Expected claim frequency = 0.0773
- Expected claim severity = 3507.04
- Expected pure premium = 3507.04 ✖ 0.0773 = **271.09**

The expected pure premium can be calculated similarly for any set of characteristics. The models are likely to underestimate the pure premium for younger drivers. For example, consider the same individual across different ages:

- At age 18, the pure premium estimate is 0.0738 ✖ 3498.188 = **258.17**
- At age 40, the pure premium estimate is 0.0852 ✖ 3526.086 = **300.42**
- At age 85, the pure premium estimate is 0.1143 ✖ 3583.844 = **409.63**

According to the fitted models, the pure premium estimate strictly increases with driver age, which is contradictory to the data, which shows that younger drivers (below 25) exhibit elevated claim frequency, and therefore higher aggregate claims.

6. Assumptions

The primary assumption for this analysis is that claim frequency is independent of claim severity. This assumption simplifies the model fitting process, enabling separate fitting of the chosen Poisson and Gamma GLMs. All applicable assumptions for fitting GLMs also apply.

7. Conclusion

The fitted Poisson and Gamma models are useful for predicting claim frequency and severity, respectively. The models identify six key risk drivers: driver age, area, vehicle age, bonus/malus, vehicle power, vehicle gas type, and region. For estimating pure premium, taking the values from the model is reasonable, but some adjustments need to be made.

The largest adjustment has to do with driver age. As discussed before, the model predicts that claim frequency strictly increases with age. However, younger drivers (18 - 25) exhibit 40.7% higher average claim frequency than drivers from age 26 to 85. This suggests that the premiums for younger drivers need to be increased compared to what the model suggests. Additionally, having a lower bonus/malus score should be rewarded with a lowered premium to encourage drivers to be more risk-averse on the road, which may lower claim frequency for all ages.

For the other risk factors, the models suggest having an elevated premium for individuals with a vehicle from brand B12 or B5, as well as higher-power vehicles.. The model also suggests having a lowered premium for diesel vehicles. For policy region, certain regions exhibit higher average claims and other regions exhibit lower average claims. This suggests that each region should be evaluated individually and the premiums should be adjusted accordingly.

8. Appendix

Full frequency model (Poisson regression):

```
                 Estimate Std. Error z value Pr(>|z|)    
(Intercept)    -3.9595806  0.0464658 -85.215  < 2e-16 ***
AreaB           0.0513736  0.0217430   2.363 0.018139 *  
AreaC           0.0861345  0.0180344   4.776 1.79e-06 ***
AreaD           0.1825332  0.0190249   9.594  < 2e-16 ***
AreaE           0.2189240  0.0203855  10.739  < 2e-16 ***
AreaF           0.1960667  0.0394062   4.976 6.51e-07 ***
VehPower        0.0134521  0.0027582   4.877 1.08e-06 ***
VehAge         -0.0387529  0.0011668 -33.213  < 2e-16 ***
DrivAge         0.0065258  0.0003941  16.557  < 2e-16 ***
BonusMalus      0.0224287  0.0003070  73.067  < 2e-16 ***
VehBrandB10    -0.0044539  0.0368129  -0.121 0.903702    
VehBrandB11     0.0745203  0.0396018   1.882 0.059871 .  
VehBrandB12     0.1542011  0.0174747   8.824  < 2e-16 ***
VehBrandB13     0.0311970  0.0409209   0.762 0.445837    
VehBrandB14    -0.1575507  0.0784414  -2.009 0.044588 *  
VehBrandB2     -0.0140297  0.0152706  -0.919 0.358232    
VehBrandB3     -0.0057063  0.0218732  -0.261 0.794185    
VehBrandB4     -0.0366273  0.0297444  -1.231 0.218172    
VehBrandB5      0.0666376  0.0247626   2.691 0.007123 ** 
VehBrandB6     -0.0468660  0.0283742  -1.652 0.098594 .  
VehGasRegular   0.0529083  0.0109391   4.837 1.32e-06 ***
RegionR21       0.1286335  0.0816174   1.576 0.115013    
RegionR22       0.0575970  0.0516137   1.116 0.264454    
RegionR23      -0.0637945  0.0607699  -1.050 0.293823    
RegionR24       0.0653959  0.0244045   2.680 0.007370 ** 
RegionR25       0.0002636  0.0447581   0.006 0.995301    
RegionR26      -0.0147385  0.0488464  -0.302 0.762857    
RegionR31      -0.1123995  0.0351734  -3.196 0.001395 ** 
RegionR41      -0.2471506  0.0455584  -5.425 5.80e-08 ***
RegionR42      -0.0201241  0.0888363  -0.227 0.820789    
RegionR43      -0.0889380  0.1340338  -0.664 0.506979    
RegionR52      -0.0128720  0.0302598  -0.425 0.670558    
RegionR53       0.0781252  0.0287355   2.719 0.006552 ** 
RegionR54      -0.0462389  0.0385950  -1.198 0.230896    
RegionR72      -0.0869893  0.0336624  -2.584 0.009761 ** 
RegionR73      -0.1245922  0.0428470  -2.908 0.003639 ** 
RegionR74       0.1823645  0.0657739   2.773 0.005561 ** 
RegionR82       0.0762450  0.0242301   3.147 0.001651 ** 
RegionR83      -0.2897689  0.0754555  -3.840 0.000123 ***
RegionR91      -0.0160456  0.0325707  -0.493 0.622267    
RegionR93      -0.0047594  0.0249970  -0.190 0.848996    
RegionR94       0.1069782  0.0669878   1.597 0.110270    
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Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 224481  on 678012  degrees of freedom
Residual deviance: 217294  on 677971  degrees of freedom
AIC: 286701
```

Since the variance of the response variable slightly exceeds the mean, a Negative Binomial GLM was also considered. The Negative Binomial model exhibits a higher AIC and performed worse than the Poisson model in terms of mean absolute error (MAE). Thus, Poisson was chosen as the final model.

```
                 Estimate Std. Error z value Pr(>|z|)    
(Intercept)    -4.5998576  0.0489300 -94.009  < 2e-16 ***
AreaB           0.0389025  0.0223437   1.741 0.081666 .  
AreaC           0.0575343  0.0185093   3.108 0.001881 ** 
AreaD           0.1288410  0.0195065   6.605 3.97e-11 ***
AreaE           0.1392386  0.0208325   6.684 2.33e-11 ***
AreaF           0.1757583  0.0406081   4.328 1.50e-05 ***
VehPower        0.0042800  0.0028213   1.517 0.129265    
VehAge         -0.0303806  0.0011699 -25.969  < 2e-16 ***
DrivAge         0.0123897  0.0004147  29.875  < 2e-16 ***
BonusMalus      0.0177170  0.0003376  52.485  < 2e-16 ***
VehBrandB10    -0.0575501  0.0377767  -1.523 0.127652    
VehBrandB11    -0.0055054  0.0407631  -0.135 0.892565    
VehBrandB12    -0.0964610  0.0180098  -5.356 8.51e-08 ***
VehBrandB13     0.0128051  0.0422121   0.303 0.761622    
VehBrandB14    -0.1908528  0.0804583  -2.372 0.017689 *  
VehBrandB2     -0.0036533  0.0157393  -0.232 0.816448    
VehBrandB3     -0.0222550  0.0225356  -0.988 0.323372    
VehBrandB4     -0.0467707  0.0306159  -1.528 0.126597    
VehBrandB5      0.0790424  0.0255692   3.091 0.001993 ** 
VehBrandB6     -0.0656771  0.0292125  -2.248 0.024560 *  
VehGasRegular   0.1038184  0.0112772   9.206  < 2e-16 ***
RegionR21       0.0661843  0.0839913   0.788 0.430702    
RegionR22       0.0747323  0.0531655   1.406 0.159827    
RegionR23      -0.3467720  0.0619069  -5.602 2.12e-08 ***
RegionR24       0.2324895  0.0250837   9.269  < 2e-16 ***
RegionR25       0.1874349  0.0461543   4.061 4.89e-05 ***
RegionR26      -0.0189938  0.0501816  -0.379 0.705059    
RegionR31      -0.1742478  0.0360886  -4.828 1.38e-06 ***
RegionR41      -0.0569804  0.0466761  -1.221 0.222176    
RegionR42       0.1314572  0.0918902   1.431 0.152548    
RegionR43      -0.1281301  0.1375043  -0.932 0.351427    
RegionR52       0.0943044  0.0311121   3.031 0.002437 ** 
RegionR53       0.3103585  0.0296317  10.474  < 2e-16 ***
RegionR54       0.0755229  0.0396956   1.903 0.057099 .  
RegionR72      -0.1388828  0.0345291  -4.022 5.77e-05 ***
RegionR73      -0.2063315  0.0438959  -4.700 2.60e-06 ***
RegionR74       0.2477684  0.0678196   3.653 0.000259 ***
RegionR82       0.1667846  0.0249722   6.679 2.41e-11 ***
RegionR83      -0.3084175  0.0770182  -4.004 6.22e-05 ***
RegionR91      -0.1453455  0.0334937  -4.339 1.43e-05 ***
RegionR93      -0.0446010  0.0257724  -1.731 0.083528 .  
RegionR94       0.0438374  0.0689713   0.636 0.525044    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for Negative Binomial(0.9728) family taken to be 1)

    Null deviance: 188765  on 678012  degrees of freedom
Residual deviance: 184737  on 677971  degrees of freedom
AIC: 281906
```

Fitted gamma model for claim severity:

```
                 Estimate Std. Error t value Pr(>|t|)    
(Intercept)     7.4618742  0.3568963  20.908  < 2e-16 ***
Exposure       -0.8955290  0.1289097  -6.947 3.82e-12 ***
AreaB           0.0645333  0.1658787   0.389  0.69725    
AreaC          -0.0948319  0.1367747  -0.693  0.48810    
AreaD          -0.0713219  0.1426763  -0.500  0.61716    
AreaE          -0.1048394  0.1510434  -0.694  0.48762    
AreaF          -0.3241517  0.3009304  -1.077  0.28142    
VehPower        0.0264675  0.0205886   1.286  0.19861    
VehAge         -0.0028532  0.0082691  -0.345  0.73007    
DrivAge         0.0003611  0.0030621   0.118  0.90614    
BonusMalus      0.0066062  0.0022198   2.976  0.00292 ** 
VehBrandB10     0.1118921  0.2478946   0.451  0.65173    
VehBrandB11     0.5233305  0.2615338   2.001  0.04540 *  
VehBrandB12    -0.0220646  0.1390693  -0.159  0.87394    
VehBrandB13     0.0213712  0.2840218   0.075  0.94002    
VehBrandB14    -0.0700710  0.5558425  -0.126  0.89968    
VehBrandB2      0.1287920  0.1078773   1.194  0.23254    
VehBrandB3      0.0422427  0.1499327   0.282  0.77814    
VehBrandB4      0.0590424  0.2054515   0.287  0.77382    
VehBrandB5     -0.1069953  0.1724457  -0.620  0.53496    
VehBrandB6     -0.1908816  0.1940795  -0.984  0.32536    
VehGasRegular   0.0994243  0.0801010   1.241  0.21453    
RegionR21       0.9482217  0.7352114   1.290  0.19716    
RegionR22       0.0672794  0.3870386   0.174  0.86200    
RegionR23      -0.1314332  0.4514840  -0.291  0.77097    
RegionR24       0.3427488  0.1827704   1.875  0.06076 .  
RegionR25       0.2094925  0.3364698   0.623  0.53354    
RegionR26       0.0382775  0.3755749   0.102  0.91882    
RegionR31      -0.0433571  0.2561297  -0.169  0.86558    
RegionR41       0.2600026  0.3332005   0.780  0.43521    
RegionR42      -0.2458225  0.6732344  -0.365  0.71501    
RegionR43       0.1324142  1.0334771   0.128  0.89805    
RegionR52      -0.0595439  0.2228680  -0.267  0.78934    
RegionR53       0.2285289  0.2173766   1.051  0.29313    
RegionR54      -0.0605018  0.2756314  -0.220  0.82626    
RegionR72       0.0174158  0.2473175   0.070  0.94386    
RegionR73      -0.1527735  0.3618803  -0.422  0.67291    
RegionR74      -0.1199605  0.4792466  -0.250  0.80235    
RegionR82       0.2411204  0.1807028   1.334  0.18210    
RegionR83      -0.3348241  0.5530993  -0.605  0.54495    
RegionR91       0.0411429  0.2490406   0.165  0.86878    
RegionR93       0.2774793  0.1888057   1.470  0.14167    
RegionR94       0.5078916  0.5715435   0.889  0.37421    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for Gamma family taken to be 39.60801)

    Null deviance: 46482  on 26443  degrees of freedom
Residual deviance: 41359  on 26401  degrees of freedom
  (195 observations deleted due to missingness)
AIC: 455576
```